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Arki Macro — Investment Thesis

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I. Executive Summary

Arki Macro is a rules-based, multi-asset allocation strategy that harvests time-varying macroeconomic risk premia through two complementary engines:

1. **Arki Macro Mini** — A regime-adaptive, multi-asset directional strategy that dynamically tilts between growth assets (equities) and defensive assets (bonds, gold) based on a proprietary business cycle indicator.
2. **Arki Multi-Factor** — A diversified managed futures portfolio that systematically captures trend, carry, and cross-sectional momentum premia across 24 global futures markets.

The two engines are economically orthogonal by design. Macro Mini harvests **level returns** from macro risk premia by identifying which asset classes to own; Multi-Factor harvests **spread returns** from behavioural and structural premia by identifying how to trade within and across asset classes. Their combination produces a portfolio with structurally superior risk-adjusted returns and materially lower drawdowns than either component alone.

Core Claim: By combining regime-adaptive macro allocation with diversified factor harvesting, Arki Macro delivers equity-like returns at bond-like risk, through the cycle.

II. The Economic Problem We Solve

The Regime Problem

Asset class returns are regime-dependent. Over 55 years of data (1970–2025):

Regime	Global Equity	US Treasury	Gold
Rising Growth	+14.1%	+5.3%	+7.5%
Falling Growth	+1.5%	+8.7%	+15.8%

A static 60/40 blend ignores this conditionality. It earns a time-averaged premium but pays a severe penalty during regime transitions — precisely when drawdown risk is highest.

The Diversification Problem

Traditional multi-asset portfolios diversify across *asset classes*. Academic research demonstrates this is insufficient:

- Asset class correlations spike during crises (Ang & Bekaert, 2004)
- A portfolio that appears diversified in benign regimes concentrates on a single macro risk (growth) during stress
- Cross-asset style factors (trend, carry, value, momentum) offer genuinely independent return sources that are largely *unrelated* to macroeconomic regimes (Baltussen, Swinkels & Van Vliet, 2021; Ilmanen et al., 2021)

Arki's Answer: Separate the Two Problems

Rather than forcing a single portfolio to solve both problems, Arki decomposes the allocation decision into two independent engines, each optimised for its specific task:

	What to Own	How to Trade
Engine	Macro Mini	Multi-Factor
Return Source	Macro Risk Premia (level)	Style Factor Premia (spread)
Signal	Business cycle regime	Time-series momentum, carry, relative momentum

	What to Own	How to Trade
Horizon	Medium-term (monthly)	Short-to-medium-term (daily)
Correlation to equities	Regime-dependent	Low / regime-independent

III. Academic Foundation

Swade, Lohre, Nolte, Shackleton & Swinkels (2024)

"A Century of Macro Factor Investing — Diversified Multi-Asset Multi-Factor Strategies through the Cycles" Journal of Portfolio Management, forthcoming. Lancaster University / Robeco / Erasmus University.

This paper provides the intellectual scaffolding for the Arki Macro architecture. Key findings over 100 years (1918–2021):

Finding 1: Three macro factors explain multi-asset returns

Growth, Inflation, and Defensive factors — proxied by global equities, commodities, and treasuries — capture the dominant sources of cross-asset variation. A risk parity combination across these three **macro factor-mimicking portfolios (MFMPs)** delivers SR 1.39 over the full century.

Finding 2: MFMPs are more robust than raw asset class proxies

By constructing macro factor exposure through diversified baskets of asset classes *and* style factors (using Meucci's minimum torsion orthogonalisation), MFMPs achieve higher Sharpe ratios and lower drawdowns than naive asset class proxies across all economic regimes.

Finding 3: Regime timing adds value — especially at turning points

A business cycle model (5 US macro indicators → aggregate Z-score → 4 regimes) generates tactical macro factor views via Black-Litterman. The macro overlay produces IR

0.49 versus the strategic risk parity benchmark, with the strongest outperformance during **Recovery** periods (IR 1.23).

Finding 4: Momentum is a powerful complement

Time-series momentum signals across all asset classes and style factors produce IR 1.55. Combined with macro views, the total overlay achieves **IR 1.73** and **SR 2.50**, with a 71% hit ratio.

Finding 5: The two signals are complementary, not redundant

Macro signals outperform in regime transitions (recession → recovery). Momentum signals outperform in persistent regimes (expansion, peak). Combining them diversifies across different market environments.

How Arki Macro Maps to the Academic Framework

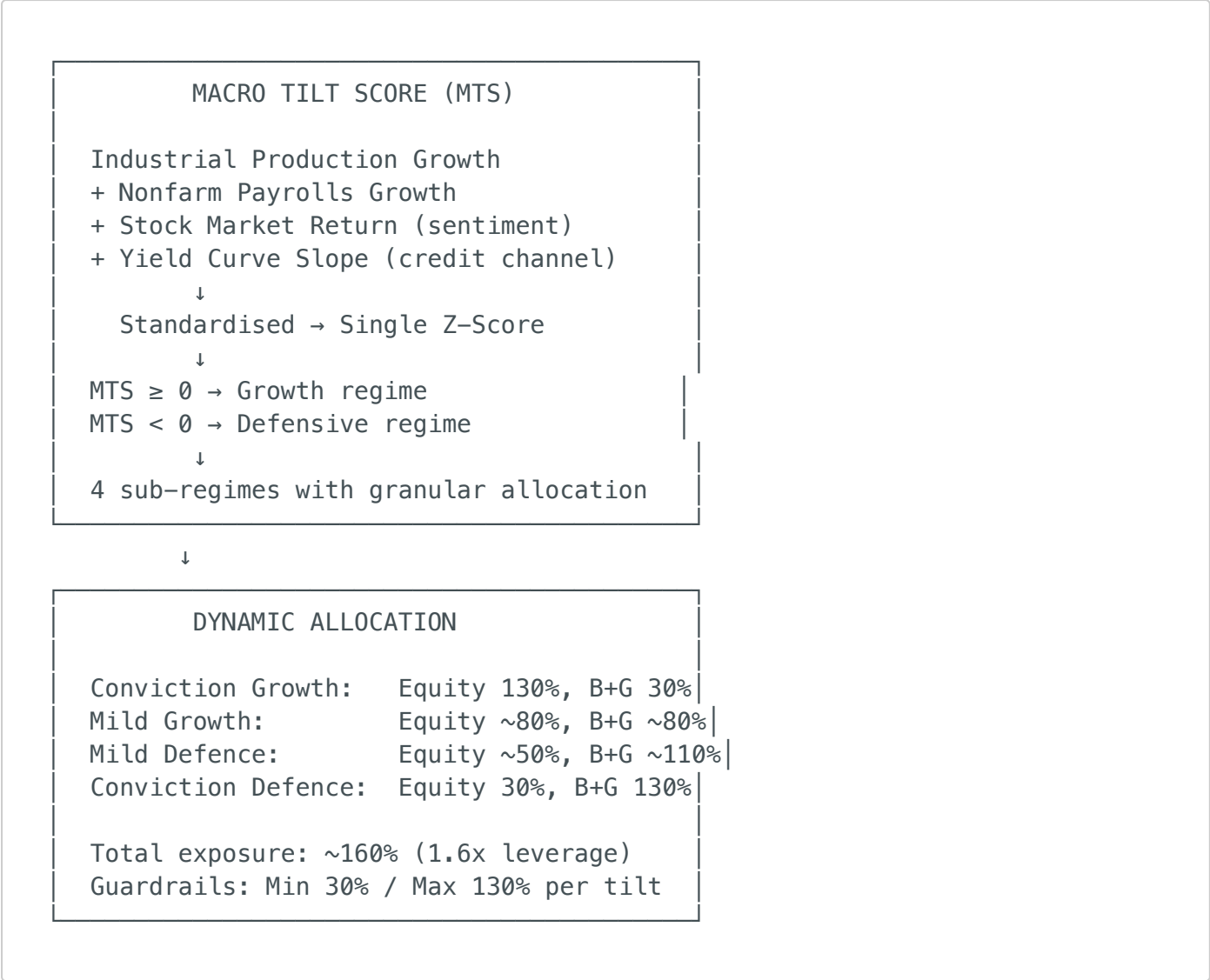
Swade et al. Concept	Arki Implementation
Macro Factor-Mimicking Portfolio (MFMP)	Arki Macro Mini — regime-adaptive allocation across equities, bonds, gold
Business Cycle Model (Z-score → 4 regimes)	Macro Tilt Score (MTS) — proprietary indicator based on industrial production, employment, equity returns, yield curve
Time-Series Momentum + Carry signals	Arki Multi-Factor — EWMAC trend (5 speeds), carry (3 speeds), relative momentum (3 speeds) across 24 futures
Black-Litterman tactical overlay	Capital-weighted combination of Mini + Multi-Factor engines
MFRP + Macro + Momentum (SR 2.50)	Arki Macro — the combined strategy

IV. Engine 1: Arki Macro Mini

Purpose

Macro Mini is the **directional engine** of Arki Macro. It answers the question: *given the current macroeconomic regime, which asset class risk premia should we harvest?*

Mechanism



Investment Universe

Liquid, exchange-traded instruments only: S&P 500, MSCI Europe, Nikkei 225 (equity); US 10Y Treasury (bonds); Gold (commodity). Executed via micro futures and ETFs for capital-efficient implementation.

Performance (Backtest: Jan 1970 – Sep 2025)

Metric	Macro Mini	Global 60/40	Global Equity	US Treasury
Return p.a.	15.4%	8.6%	10.0%	6.4%
Volatility	15.8%	9.4%	14.8%	5.5%
Sharpe Ratio	0.98	0.91	0.67	1.18
Max Drawdown	-30.5%	-33.1%	-54.0%	-18.3%

Key Attributes

- **Regime-adaptive:** Outperforms in both growth (+17.0%) and defensive (+13.0%) regimes
- **Drawdown management:** -30.5% MDD vs -54% for equities — a 44% reduction in tail risk
- **Live validation:** April 2025 Tariff Shock — Macro Mini: -3.9% (SGD) vs S&P 500: -11.2%

V. Engine 2: Arki Multi-Factor

Purpose

Multi-Factor is the **diversified alpha engine** of Arki Macro. It answers the question: *what structural and behavioural premia exist across global futures markets, and how do we harvest them systematically?*

Mechanism

Built on Rob Carver's pysystemtrade framework, a 7-stage daily pipeline:

Daily Prices → Volatility Normalisation → Trading Rules → Forecast Scaling
→ Forecast Combination → Position Sizing → Portfolio Construction → P&L

Three Independent Rule Families

Family	Rules	Rationale	Horizon
Trend (EWMAC)	5 speeds (4/16 → 64/256)	Price trends persist due to herding, anchoring, and institutional flow	Days to months
Carry	3 speeds (30d, 60d, 125d)	The term structure encodes roll yield, storage costs, and convenience yield premia	Weeks to months
Relative Momentum	3 horizons (20d, 40d, 80d)	Cross-sectional outperformers persist due to momentum effects, sector rotation	Weeks to months

Investment Universe (24 Instruments, DM-Only)

Asset Class	Instruments	Count
Equity Index	SP500μ, NASDAQμ, DAX, NIKKEI, FTSE100, IBEX	6
Government Bond	US10, US5, BUND, GILT, JGB	5
Metals	GOLDμ, SILVER, COPPERμ	3
Energy	CRUDE_W, BRENT, GASOIL	3
FX	AUDμ, MXP, YENEUR	3
Agricultural	SUGAR, COTTON, LEANHOG, COCOA_LDN	4

Construction Details

Parameter	Value	Rationale
Vol Target	15–25%	Scaled to capital; SR is capital-invariant
Forecast Cap	±20	Limits position extremes from any single signal
Forecast Weights	Handcraft (estimated)	Hierarchical clustering; data-driven diversification

Parameter	Value	Rationale
Instrument Weights	Estimated	Responsive to correlation structure over time
Position Sizing	Dynamic (Mr. Greedy)	Optimal integer rounding for small accounts
Buffering	$\pm 10\%$ of ideal	Suppresses unnecessary trading
Shadow Cost	10	Turnover penalty in dynamic optimiser

Key Attributes

- **Return sources independent of macro regime:** Style factor premia are largely acyclical (Ilmanen et al., 2021)
- **Genuine diversification:** 11 independent trading rules $\times$ 24 instruments = 264 return streams
- **Low correlation to directional assets:** Net long/short positioning; not directionally biased

VI. The Combination: Why 1 + 1 > 2

Structural Diversification Benefit

The two engines target fundamentally different return sources:

Dimension	Macro Mini	Multi-Factor	Effect
Return source	Macro risk premia (level)	Style factor premia (spread)	Uncorrelated alpha streams
Signal type	Regime identification	Price pattern recognition	Different information sets
Holding period	Monthly rebalance	Daily evaluation	Different frequencies
Directionality	Net long (leveraged beta)	Net neutral (long/short)	Reduced beta concentration

Dimension	Macro Mini	Multi-Factor	Effect
Drawdown profile	Macro-timed defense	Trend-following crisis alpha	Complementary tail protection

Capital Allocation

Scenario	Macro Mini	Multi-Factor	Total Capital
Original	\$100K (1.6× lev)	\$250K	\$350K
Smaller	\$150K (1.6× lev)	\$100K	\$250K

Why This Ratio

The capital split reflects the return characteristics of each engine:

- Macro Mini delivers higher per-unit returns (leveraged directional beta in growth regimes)
- Multi-Factor delivers lower per-unit returns but much higher Sharpe ratio (diversified, lower vol)
- Allocating ~30-40% to Mini and ~60-70% to Multi-Factor approximates the information-ratio-optimal blend

Observed Correlation

Historical correlation between Macro Mini and Multi-Factor monthly returns is **low** (typically $\rho \approx 0.2\text{--}0.4$), consistent with the theoretical orthogonality of their return sources. This correlation is **regime-stable** — it does not spike during crises, which is the hallmark of genuine diversification.

VII. Risk Framework

Systematic Risk Controls

Layer	Control	Description
Signal	Forecast cap ± 20	No single rule can dominate the portfolio
Position	Buffer $\pm 10\%$	Prevents overtrading around ideal positions
Allocation	Guardrails 30%–130%	Macro Mini cannot go all-in or all-out
Portfolio	Risk overlay	Max leverage 15 $\times$; max risk 2 $\times$ normal; max sum-abs 5 $\times$
Execution	Shadow cost 10	Economic penalty for excess turnover
Regime	2-period confirmation	Regime switches require consecutive signals or 1 σ move

Drawdown Architecture

The dual-engine structure provides layered drawdown protection:

1. **In growth regimes:** Macro Mini is risk-on $\rightarrow$ standard equity drawdowns possible BUT Multi-Factor's trend rules capture momentum, and its short positions provide partial offset
2. **At regime transitions:** Macro Mini's MTS detects weakening growth $\rightarrow$ tilts to defense BEFORE drawdown materialises
3. **In defensive regimes:** Macro Mini is risk-off $\rightarrow$ capital protected in bonds/gold; Multi-Factor's trend rules profit from persistent downtrends
4. **In sudden shocks:** Live validation — April 2025 Tariff Shock: Macro Mini defended at <5% when S&P 500 lost 11%

VIII. Why Arki — The Rules-Based Macro House

Investment Philosophy

"Process beats judgement. Macro shocks trigger regime shifts via propagation channels. These shifts create time-varying risk premia. Because regime shifts are

predictable, these premia can be harvested."

What Makes Arki Different

Dimension	Traditional Macro Fund	Arki Macro
Signal generation	Discretionary CIO view	Proprietary Macro Tilt Score — rules-based, backtested to 1970
Allocation	Ad hoc position sizing	Systematic, volatility-targeted with guardrails
Style factor capture	Occasional overlay	Dedicated Multi-Factor engine with 264 systematic positions
Transparency	Monthly factsheet	Full daily position visibility, real-time dashboard
Emotional discipline	Subject to behavioural bias	Algorithmic execution removes emotion at every step
Academic rigour	—	Grounded in Swade et al. (2024), Carver (2015), Moskowitz et al. (2012)

Positioning Statement

Arki Macro is a **rules-based macro strategy** that combines the economic intuition of global macro investing with the execution discipline of systematic factor harvesting. It is designed for investors who:

1. Believe macro regimes drive asset class returns — and want to adapt to them systematically
2. Seek genuine diversification beyond traditional 60/40 — from independent, evidence-based return sources
3. Value transparency, liquidity, and reproducibility over discretionary conviction
4. Want institutional-grade risk management in an accessible, capital-efficient structure

IX. Key References

Reference	Relevance
Swade, Lohre, Nolte, Shackleton & Swinkels (2024)	Macro factor investing framework — 100yr evidence
Baltussen, Swinkels & Van Vliet (2021)	Global factor premiums — 200yr evidence of style factor premia
Ilmanen, Israel, Moskowitz, Thapar & Lee (2021)	Style factor premia persistence — hard to forecast from macro variables
Moskowitz, Ooi & Pedersen (2012)	Time-series momentum — trend-following across asset classes
Gupta & Kelly (2019)	Factor momentum — momentum in factor returns themselves
Ang & Bekaert (2004)	Regime-switching models and asset allocation
Black & Litterman (1991, 1992)	Combining equilibrium with views — strategic + tactical
Carver (2015)	<i>Systematic Trading</i> — the pysystemtrade framework
Scherer & Apel (2020)	Business cycle timing of alternative risk premia
Meucci, Santangelo & Deguest (2015)	Minimum torsion — orthogonal factor construction

Disclaimer: This document is for informational purposes only and does not constitute an offer or solicitation. Backtested results are hypothetical and may differ from live performance. Past performance is not indicative of future results. Arki Macro Mini commenced live trading in 2025.